

AMTI (NMTC) - 2011
BHASKARA CONTEST - FINAL - JUNIOR LEVEL

1. If $a = 2011^{2010}$, $b = 2010^{2011}$, $c = (2010 + 2011)^{2010 + 2011}$, and $d = 2011$, find the value of $\frac{bc(a+d)}{(a-b)(a-c)} + \frac{ac(b+d)}{(b-a)(b-c)} + \frac{ab(c+d)}{(c-a)(c-b)}$.
2. The internal bisectors of angles A, B, C of triangle ABC meet the circumcircle respectively at P, Q, R. I is the incentre of ABC. PQ meets BC, CI and CA at T, Y, L respectively PR meets BC, BI and AB at S, X, M respectively.
Prove :
(i) I is the orthocenter of the triangle PQR.
(ii) RQTS, RQYX and RQLM are cyclic quadrilaterals.
3. Let $A = \{a^2 + 4ab + b^2 \mid a, b \text{ are positive integers}\}$. Prove that $2011 \neq A$.
4. If $1 \leq x \leq 64$, find the greatest value of the expression. $(\log_2 x)^4 + 12(\log_2 x)^2 \log_2 \left(\frac{8}{x}\right)$.
5. If perpendiculars are drawn from the vertices of a square to a line in the plane of the square (the line is not parallel to any side or diagonal), prove that the sum of the squares of the perpendiculars from one pair of opposite vertices exceeds twice the product of the perpendiculars from the other pair of opposite vertices by the area of the square.
6. x, y, z are real numbers such that $x + y + z = 3$ and $xy + yz + zx = a$ (where a is a real parameter). Determine the value of 'a' for which the difference between the maximum and minimum possible value of x is equal to 8.
7. abc is a three digit number. ab, bc, ca are two digit numbers. Determine all three digit numbers abc such that $abc = ab + bc + ca$.
8. If a, b, c, d are four real numbers such that $a + 2b + 3c + 4d \geq 30$, prove that $a^2 + b^2 + c^2 + d^2 > 30$.

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